



Characteristics of a Centrifugal Pump

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Objective

Determine the performance characteristics of a centrifugal pump by measuring flow rate, pressures, and torque. Construct dimensional and nondimensional performance curves, evaluate efficiency, and verify similarity using homologous coefficients.

Key Equations

Input Power: $P_{in} = FL\omega$

Output Power: $P_{out} = \gamma Q h_p$

Efficiency: $\eta = P_{out}/P_{in}$

Pump Head: $h_p = p_2 - p_1 / \gamma + V_2^2 / 2g - V_1^2 / 2g + (z_2 - z_1)$

*** If D doubles: $Q = D^3 \rightarrow 8Q$

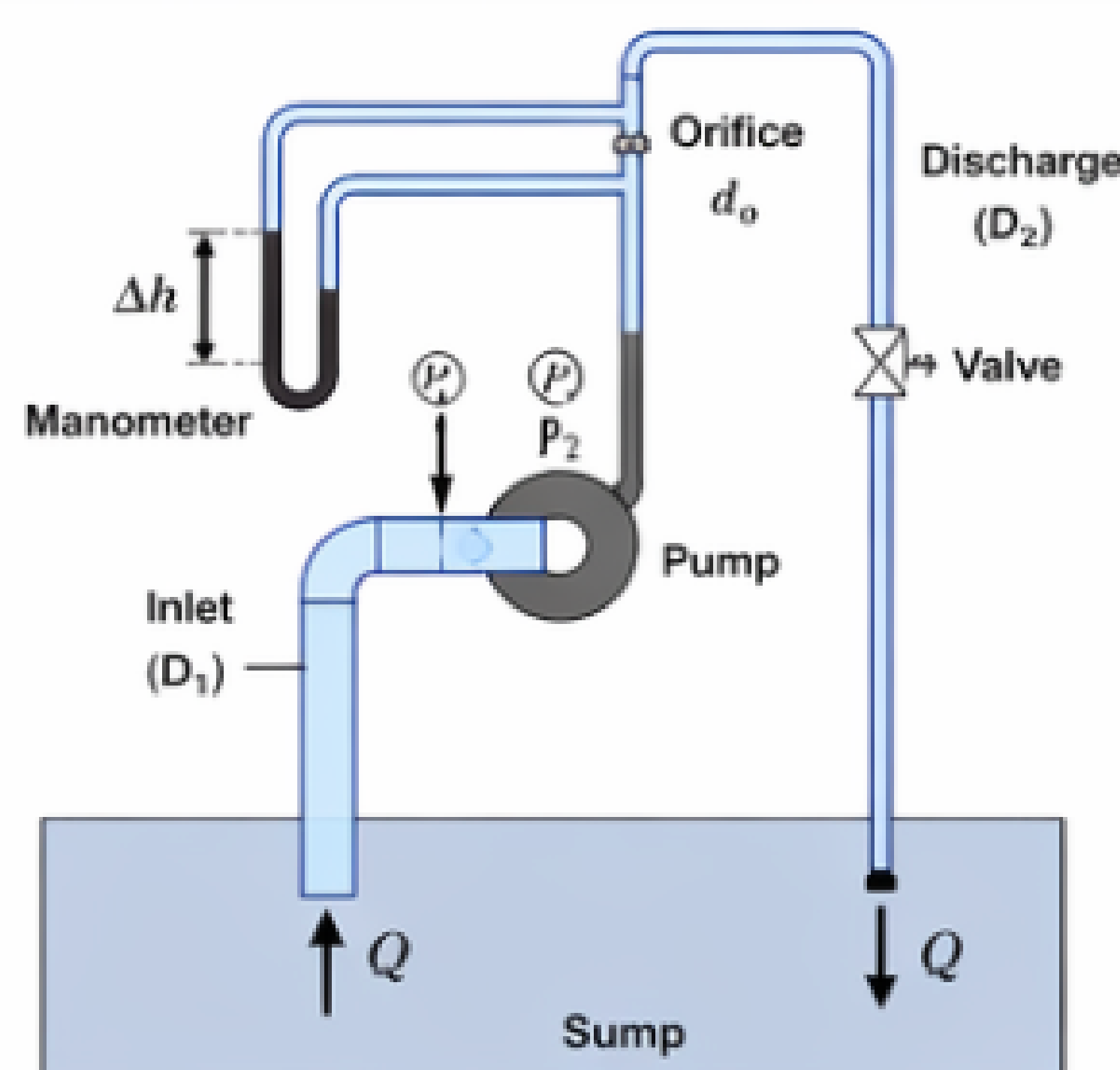
Dimensionless Coefficients

$C_Q = Q / (\omega D_{imp}^3) \rightarrow$ normalized flow rate

$C_P = P_{in} / (\rho \omega^3 D_{imp}^5) \rightarrow$ normalized power

$C_H = g h_p / (\omega^2 D_{imp}^2) \rightarrow$ normalized head (energy)

Experimental Setup



Sample Calculations

Flow rate, $Q = 0.01724 \text{ m}^3/\text{s}$

Velocities, V_1 & $V_2 = 3.62$ & 8.31 m/s

Pump head, $h_p = 20.58 \text{ m}$

Input power, $P_{in} = 6385 \text{ W}$

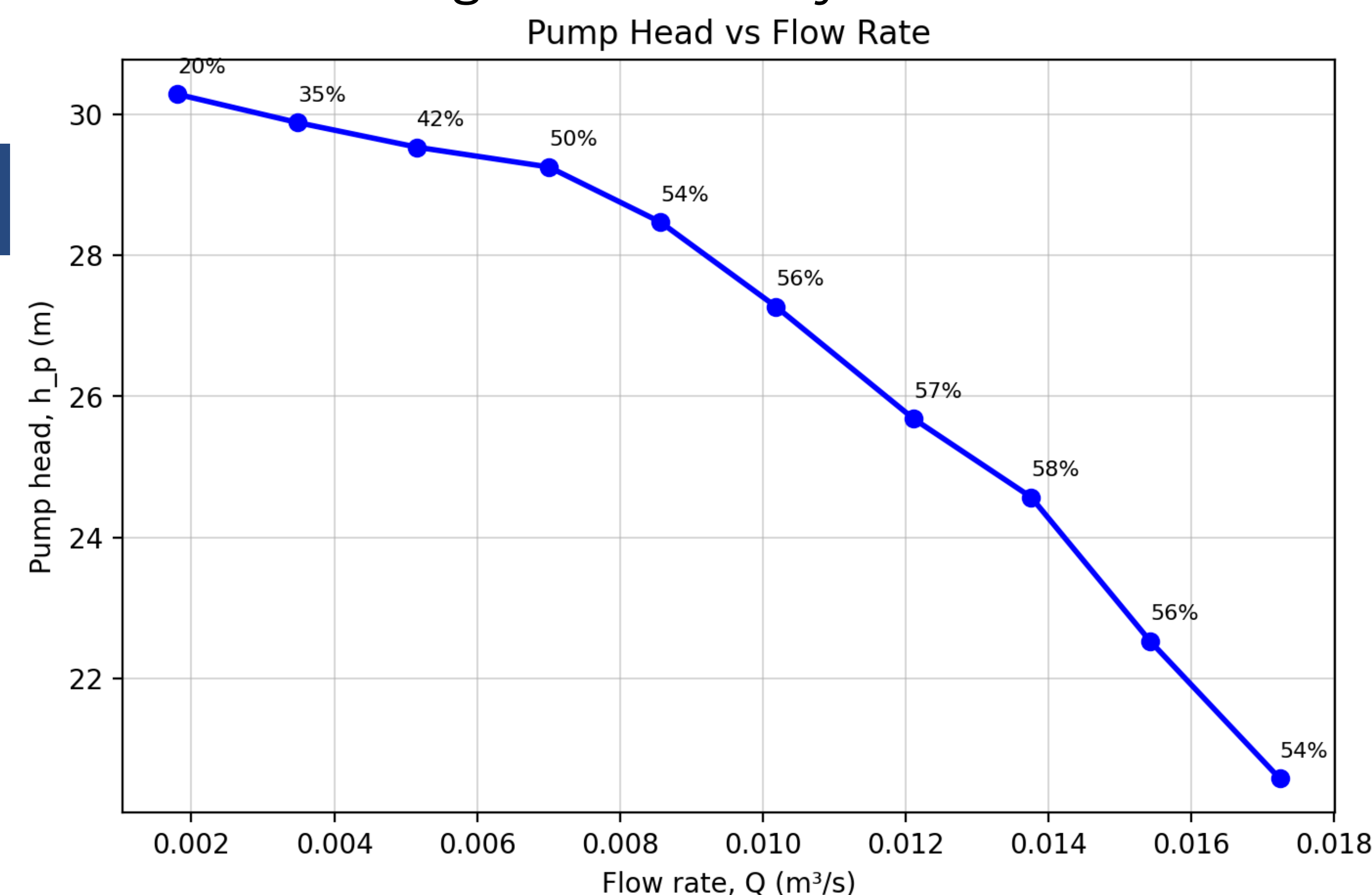
Output power, $P_{out} = 3479 \text{ W}$

Efficiency, $\eta = 0.545$ (54.5%)

Max vertical height, $h_{p,max} \sim 30 \text{ m}$

Key Results

- Maximum efficiency $\sim 58\%$ at $Q \sim 0.0172 \text{ m}^3/\text{s}$
- Manufacturer: 69% at $Q = 0.0172 \text{ m}^3/\text{s}$
- h_p decreases: 30.3 to 20.6 m , as Q increases
- Measured efficiency is lower due to losses and non-ideal effects
- P_{in} increases slightly with flow rate, while P_{out} increases to a maximum and then decreases
- Efficiency $\geq 75\%$ of max occurs for: $Q \approx 0.010 - 0.015 \text{ m}^3/\text{s} \rightarrow$ good versatility



Discussion

- Efficiency increases with flow rate, reaches a maximum (BEP), then decreases at higher Q rates
- Energy losses occur due to pipe friction, turbulence, mechanical friction and leakage: $P_{in} = P_{out} + P_{lost}$
- Pump head is due to pressure rise
- Nondimensional curves collapse to single trends

Conclusions

- Pump efficiency peaks at roughly 58%
- Pump head decreases as flow rate increases
- Pressure rise is dominant contributor to head
- Nondimensional analysis supports pump similarity theory
- Homologous curves consistent within deviations
- Both graphs show peak efficiency at same Q , while head decreases and power exhibits a peak behavior

